Ofqual QAN	**600/6407/9 / 603/0080/2**
Quartz ID	**1047 / 1125**
Qualification	**Level 3 Diploma in Computing / Level 3 International Foundation Diploma for Higher Education Studies**

Assessment	**Introduction to Computer Science**
Assessment Type	**Examination**
Date	**Tuesday 11th March 2025**
Time	**09.00am-12.00pm**

Examination Paper

Answer ALL questions.

Clearly cross out any surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Section A – Multiple Choice Question
Answer these in your answer booklet – do NOT write on this Question Paper.
Write ONE (1) correct answer from A, B, C, or D for each question in your answer booklet.
Each question is worth 1 mark.

Question 1

What does CPU stand for?

A Computing Processing Unit

B Central Processing Unicode

C Chip Processing Unit

D Central Processing Unit

1 mark

Question 2

Which of these home considerations can **NOT** be controlled remotely using Smart Technology?

A Room lighting

B Heating

C Security alarms & cameras

D Wi-Fi speed

1 mark

Question 3

Which type of network is best for secure connections over a public network?

A PAN

B WPAN

C VPN

D MAN

1 mark

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Marks

Question 4

What is the main advantage of using hexadecimal instead of binary for data representation?

A Hexadecimal is easier for computers to process.

B Hexadecimal allows data to be represented in fewer digits.

C Hexadecimal is compatible with ASCII.

D Hexadecimal eliminates the need for encryption.

1 mark

Question 5

What type of network connects computers within a single building or campus?

A WAN

B LAN

C MAN

D VPN

1 mark

Question 6

Which type of compression reduces file size by removing some data permanently?

A Lossless compression

B Virtual compression

C Lossy compression

D Archive compression

1 mark

Question 7

Which input device category does a fingerprint scanner typically belong to?

A Key and button devices

B Pointing devices

C Visual-digital devices

D Biometric devices

1 mark

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Marks

Question 8

Which gate inverts the input signal?

A NOT

B OR

C NAND

D AND

1 mark

Question 9

Which of the following is the primary purpose of FireWire?

A Charging devices

B Wireless communication

C High-speed data transfer

D Displaying output

1 mark

Question 10

What does bit depth in digital sound representation determine?

A The number of samples per second.

B The range of possible values for each sample.

C The duration of the sound file.

D The compression type used.

1 mark

Question 11

Which of the following is **NOT** one of the four basic functions of a computer system?

A Input

B Processing

C Storage

D Networking

1 mark

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Question 12

How much memory can a computer with a 32-bit address bus address?

A 2 GB

B 4 GB

C 64 GB

D 128 GB

1 mark

Question 13

Why might a user choose to install a dedicated sound card?

A To improve graphics performance

B To enable internet access

C To achieve higher-quality audio output

D To increase overall storage capacity

1 mark

Question 14

Which device allows artists and designers to create digital sketches using a stylus on a sensitive surface?

A Graphic Tablet

B Touchscreen

C Barcode Scanner

D MIDI Controller

1 mark

Question 15

Which component stores configuration information about a computer's hardware?

A CPU

B Cache

C USB Port

D CMOS

1 mark

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Marks

Question 16

Which encryption protocol ensures data transmission security for websites?

A SSL/TLS

B ASCII

C SFTP

D UTF-8

1 mark

Question 17

What type of license allows multiple users on a network to access a single copy of software?

A End User License Agreement

B Proprietary license

C Site license

D Workstation license

1 mark

Question 18

Which of the following combinations represents a ***NAND*** gate?

A OR gate followed by a NOT gate

B NOT gate followed by an AND gate

C AND gate followed by a NOT gate

D NOT gate followed by an OR gate

1 mark

Question 19

Which component of the CPU is responsible for performing calculations?

A Control Unit (CU)

B Immediate Access Store (IAS)

C Cache Memory

D Arithmetic and Logic Unit (ALU)

1 mark

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Marks

Question 20

Which of these is a common sound file format for lossy compression?

A FLAC

B WAV

C MP3

D PCM

1 mark

Question 21

Which is an example of computer storage that is temporary?

A ROM

B RAM

C DVD

D Hard Drive

1 mark

Question 22

Which topology connects all devices to a central hub or switch?

A Bus

B Ring

C Star

D Mesh

1 mark

Question 23

What makes Raspberry Pi and single-board computers popular for educational purposes?

A Their compact and affordable design

B Their ability to handle complex quantum calculations

C Their integration into industrial machines

D Their exclusive use in advanced scientific research

1 mark

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Marks

Question 24

Which of the following utility software is primarily used to restore files after a computer crashes?

A Disk defragmenter

B Data recovery utility

C File compression utility

D Antivirus software

1 mark

Question 25

Which of the following is **TRUE** about bitmap images?

A They use geometrical objects for representation.

B They are ideal for scalable graphics.

C They are composed of mathematical formulae.

D They lose quality when resized

1 mark

Question 26

Which of the following is an example of an output device for tactile feedback?

A Monitor

B Speakers

C Multimedia projector

D Haptic feedback device

1 mark

Question 27

Which of the following best describes the purpose of two's complement?

A To represent fractional numbers

B To represent negative integers in binary

C To simplify hexadecimal conversion

D To perform binary addition

1 mark

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Marks

Question 28

What does HTML stand for?

A Hyperlink Text Markup Language

B Hypertext Transfer Management Language

C Hypertext Markup Language

D Hyperlink Transfer Monitoring Language

1 mark

Question 29

Which of the following is **NOT** an example of an operating system?

A Windows

B Linux

C Excel

D iOS

1 mark

Question 30

How is the number 12 represented in binary?

A 1100

B 1010

C 1110

D 1001

1 mark

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Marks

SECTION B
ANSWER ALL QUESTIONS

Question 31

a) A computer system at a holiday coach company monitors how many hours coach drivers have worked in a day.

The system uses satellite navigation to track the coach journeys and sends a digital alert to the drivers, via a fitted navigation device, telling them when they are legally required to take a break.

i) State **ONE (1)** output device you would expect to find as part of the navigation device. **1**

ii) Identify and describe **ONE (1)** problem with relying on satellite navigation to record the coach journey. **2**

iii) The drivers used to track the journeys themselves and send a SMS to an office worker who would record all the data on paper. **2**

State and describe **ONE (1)** advantage of the new computerised sat-nav system.

iv) Drivers must take a break after every 4 hours or 100 miles travelled. **3**

Draw a logic gate diagram that represents the two input checks (H – hours done) and (M – miles travelled) and the output (B - requires break)

b) Describe **TWO (2)** purposes of a motherboard. **2**

Total 10 Marks

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Marks

Question 32

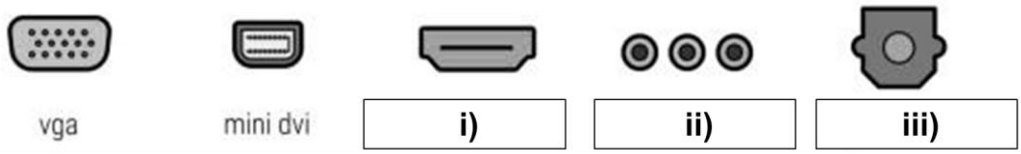
a) Developments in processors have led to improved computing performance.

i) Explain **TWO (2)** ways in which multiple cores in a processor improve performance. **2**

ii) State **ONE (1)** other feature of a processor that improves performance. **1**

b) The diagram below shows a range of ports that could be found on a desk-top personal computer. **3**

Complete the names of the missing ports, i), ii) and ii). Two examples are already given.



c) Nat is buying a new printer for her business. Discuss **TWO (2)** advantages of buying an ink jet printer **and TWO (2)** advantages of a laser printer. **4**

Total 10 Marks

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Marks

Question 33

a) A company plans to expand its operations and needs to choose a new operating system (OS).

Their options include a closed-source proprietary OS (e.g., Windows Server) or an open-source OS (e.g., Ubuntu).

i) Explain **TWO (2)** purposes of an operating system. **2**

ii) Describe **TWO (2)** benefits of choosing a closed-source OS. **2**

iii) Describe TWO **(2)** benefits of choosing an open-source OS. **2**

b) Explain the term ‘firmware’, **and** state why it is important. **2**

c) State the type of software user interface being described below:

i) Relies on the user entering text to issue instructions. **1**

ii) Uses windows, icons, menus and pointers for user interaction. **1**

Total 10 Marks

Question 34

a) Explain **TWO (2)** differences between primary and secondary storage, providing an example of each. **4**

b) Describe **TWO (2)** advantages **and TWO (2)** disadvantages of optical storage media. **4**

c) Define the term ‘encryption’ **and** explain how it enhances data security? **2**

Total 10 Marks

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Marks

Question 35

a) For each process below, describe the steps taken:

i) The process of adding **TWO (2)** 3-bit binary numbers, using an example. **3**

ii) Explain the steps for converting the decimal number 26 into hexadecimal. **3**

b) Describe how sound is represented in a computer system **and** state **TWO (2)** factors that affect digital audio quality. **3**

c) Explain the logical difference in outputs between an OR gate **and** a NOR gate. **1**

Total 10 Marks

Question 36

a) Explain **TWO (2)** advantages **and TWO (2)** disadvantages of peer-to-peer (P2P) networks. **4**

b) Explain the term ‘Social Engineering’ **and** give an example. **2**

c) Describe the roles of HTML, CSS **and** JavaScript in creating a website. **3**

d) Name the security threat where hackers use software that can automatically try all combinations of passwords. **1**

Total 10 Marks

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Marks

Question 37

a) Many people are *ethically* concerned with the growth of *AI* and *social media*.

i) State what is meant by the phrase '*ethically* concerned'. **1**

ii) State **ONE (1)** reason why people are concerned with the growth of *social media*. **1**

iii) State what is meant by the term AI. **1**

iv) State **TWO (2)** concerns people have with the growth of AI. **2**

b) "Virtual Reality is a positive development in Computing" **5**

Discuss whether you agree with this statement, include in your answer:

- possible ethical issues
- the impact on society
- the Digital Divide.

Total 10 Marks

End of paper

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